

Connor Pink

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EDUCATION

University of Guelph

Guelph, ON

Master of Science in Computer Science – AI and Machine Learning Focus

Jan. 2025 – Present

- Research: Computer Vision, Generative AI, and Explainable AI in Healthcare. Coursework: Deep Learning, Advanced Machine Learning

Trent University

Peterborough, ON

Bachelor of Science (Honours) in Computer Science with Co-op – 84.08% Average

Sept. 2020 – July 2024

EXPERIENCE

Graduate Research and Teaching Assistant

Jan. 2025 – Present

Guelph University

Guelph, ON

Research Assistant – Dr. Sykes AI and Healthcare Lab

- Develop Python and PyTorch software for generative and computer vision models in healthcare imaging, training diffusion and GAN architectures for synthetic data generation
- Build end-to-end pipelines for image segmentation and model evaluation, creating reproducible experimentation and verification workflows across data preparation, training, and analysis
- Deploy proof-of-concept AI systems with Docker on GCP and AWS for scalable inference, while applying explainable AI methods to inspect model behaviour and support interpretation
- Contribute to academic papers and present technical findings to cross-functional collaborators, translating open-ended research problems into actionable engineering next steps

Teaching Assistant

- Support CIS 3110 Operating Systems, CIS 4150 Software Testing, and CIS 3120 Digital Systems by leading labs, grading assignments, and clearly communicating complex technical material during office hours
- Guide students through debugging, test design, and systems concepts, reinforcing rigorous reasoning about software behaviour and hardware-software interaction

PROJECTS

Generative AI Pipeline for Medical Imaging | *Python, PyTorch, Diffusion Models*

Jan. 2025 – Present

- Designed and deployed an end-to-end generative AI pipeline for synthetic medical image generation with reproducible experimentation workflows
- Implemented fine-tuning and evaluation workflows on healthcare datasets, achieving a 15% quality improvement through iterative validation and error analysis
- Containerized inference services with Docker and deployed them to GCP, improving reproducibility and scalability for research use
- Applied explainable AI techniques to inspect model outputs and communicate system limitations to collaborators

ML-Based Bug Detection System | *Python, Machine Learning, Data Analysis*

Jan. 2025 – July. 2025

- Partnered with Guelph's industry sponsor Nevvon to investigate application-specific bugs and frame the problem as a data-driven ML analysis task
- Built Python-based workflows to detect, classify, and diagnose bug patterns, supporting root-cause analysis and prioritization in a production environment
- Delivered actionable findings and documented evaluation results for the development team, contributing to improved application stability and faster debugging

TECHNICAL SKILLS

Languages: Python, Java, C, C#, JavaScript, TypeScript, PL/SQL, PHP, HTML/CSS, PowerShell

AI/ML: PyTorch, Monai, WandB, TensorFlow, Scikit-learn, Diffusion Models, GANs, U-Net, LangChain, OpenAI API

Validation & Analysis: Model Evaluation, Explainable AI, Data Analysis, Experimental Design, Debugging, Reproducible Experimentation

Systems & Tools: Docker, GCP, AWS, Azure, Git, GitHub, GitHub Actions, CI/CD Pipelines, Kubernetes, Azure DevOps, Jupyter, Conda, VS Code

Software & Data: RESTful APIs, React, Node.js, Express, .NET, MongoDB, MySQL, Technical Writing